

$$\begin{aligned} & \xrightarrow{(22)} \frac{1}{16\pi^2} \\ & \left(-\frac{1}{54} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_d^{\mathbf{p}}] + \frac{5}{144} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_d^{\mathbf{p}}] - \frac{2}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_d^{\mathbf{p}}] - \frac{1}{18} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_e^{\mathbf{p}}] + \right. \\ & \quad \frac{5}{48} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^{\mathbf{p}}] - \frac{2}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_e^{\mathbf{p}}] + \left(\frac{1}{4} \mathbf{g}_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} - \frac{1}{144} \sum_{\mathbf{p}} (4 \mathbf{g}_1^4 + 9 \mathbf{g}_2^4) \right) \\ & \quad \text{LF}_{3,0}[\mathbf{m}_l^{\mathbf{p}}] + \frac{1}{96} (-24 \mathbf{g}_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} (5 \mathbf{g}_1^4 + 6 \mathbf{g}_2^4)) \text{LF}_{4,-1}[\mathbf{m}_l^{\mathbf{p}}] - \\ & \quad \frac{1}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_l^{\mathbf{p}}] + \left(\frac{3}{4} \mathbf{g}_2^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} - \frac{1}{432} \sum_{\mathbf{p}} (4 \mathbf{g}_1^4 + 81 \mathbf{g}_2^4) \right) \text{LF}_{3,0}[\mathbf{m}_q^{\mathbf{p}}] + \\ & \quad \left(-\frac{3}{4} \mathbf{g}_2^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \frac{1}{288} \sum_{\mathbf{p}} (5 \mathbf{g}_1^4 + 54 \mathbf{g}_2^4) \right) \text{LF}_{4,-1}[\mathbf{m}_q^{\mathbf{p}}] - \frac{1}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_q^{\mathbf{p}}] - \\ & \quad \frac{2}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_u^{\mathbf{p}}] + \frac{5}{36} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_u^{\mathbf{p}}] - \frac{8}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_u^{\mathbf{p}}] - \frac{1}{36} \mathbf{g}_1^4 \text{LF}_{3,0}[\tilde{\mu}] - \\ & \quad \frac{1}{24} \mathbf{g}_1^4 \text{LF}_{4,-1}[\tilde{\mu}] + \frac{2}{45} \mathbf{g}_1^4 \text{LF}_{5,-2}[\tilde{\mu}] - \frac{1}{32} \mathbf{g}_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{16} \mathbf{g}_1^4 \mathbf{m}_1^2 \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] - \\ & \quad \frac{17}{32} \mathbf{g}_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{16} \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + \mathbf{g}_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] - \\ & \quad \frac{3}{4} \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{2} \mathbf{g}_2^4 \text{LF}_{3,3,-3}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{3,3,-2}[\mathbf{m}_2, \tilde{\mu}] - \\ & \quad \frac{1}{2} \mathbf{g}_2^4 \text{LF}_{4,2,-3}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{4,2,-2}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{6} \mathbf{g}_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] + \\ & \quad \frac{1}{12} \mathbf{g}_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \frac{1}{6} \mathbf{g}_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{p}}] + \\ & \quad \frac{1}{12} \mathbf{g}_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{p}}] + \frac{1}{6} \mathbf{g}_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] + \\ & \quad \frac{1}{6} \mathbf{g}_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] + \frac{1}{8} \overline{a_e^{\text{pr}}} a_e^{\text{pr}} (-3 \mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{4,1,-1}[\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] + \\ & \quad \frac{1}{6} \mathbf{g}_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] - \frac{1}{4} \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{2,1,0}[\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_l^{\mathbf{s}}] + \\ & \quad \frac{1}{4} \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{3,1,-1}[\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_l^{\mathbf{s}}] + \frac{1}{6} \mathbf{g}_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] + \\ & \quad \frac{1}{6} \mathbf{g}_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] + \frac{1}{8} \overline{a_d^{\text{pr}}} a_d^{\text{pr}} (-5 \mathbf{g}_1^2 + 3 \mathbf{g}_2^2) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] + \\ & \quad \frac{1}{2} \mathbf{g}_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] - \frac{3}{4} \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_q^{\mathbf{s}}] + \\ & \quad \frac{3}{4} \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_q^{\mathbf{s}}] + \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] - \\ & \quad \frac{3}{4} \mathbf{g}_2^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] - \frac{1}{24} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (\mathbf{g}_1^2 - 9 \mathbf{g}_2^2) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] + \\ & \quad \frac{3}{4} \mathbf{g}_2^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] - \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \\ & \quad \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \frac{1}{4} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] + \\ & \quad \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \frac{1}{16} \mathbf{g}_1^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\ & \quad \frac{1}{16} \mathbf{g}_1^4 \mathbf{m}_1^2 \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] + \frac{5}{24} \mathbf{g}_1^4 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] + \frac{1}{16} \mathbf{g}_1^4 \text{LF}_{4,2,-3}[\tilde{\mu}, \mathbf{m}_1] - \\ & \quad \frac{1}{16} \mathbf{g}_1^4 \mathbf{m}_1^2 \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{6} \mathbf{g}_1^4 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] + \frac{3}{16} \mathbf{g}_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \\ & \quad \frac{1}{16} \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{3,$$